

Ping_blocked

```
mininet> h1 ping h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
From 10.0.0.1 icmp_seq=1 Destination Host Unreachable
From 10.0.0.1 icmp_seq=2 Destination Host Unreachable
From 10.0.0.1 icmp_seq=3 Destination Host Unreachable
From 10.0.0.1 icmp_seq=4 Destination Host Unreachable
From 10.0.0.1 icmp_seq=5 Destination Host Unreachable
From 10.0.0.1 icmp_seq=6 Destination Host Unreachable
From 10.0.0.1 icmp_seq=7 Destination Host Unreachable
From 10.0.0.1 icmp_seq=8 Destination Host Unreachable
From 10.0.0.1 icmp_seq=9 Destination Host Unreachable
^C
--- 10.0.0.3 ping statistics ---
11 packets transmitted, 0 received, +9 errors, 100% packet loss, time 10227ms
pipe 4
```

Ping_allowed

```
mininet> h2 ping h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=4.61 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=1.55 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.143 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.054 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=0.222 ms
^C
--- 10.0.0.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4034ms
rtt min/avg/max/mdev = 0.054/1.315/4.608/1.735 ms
```

Pingall

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 X
h2 -> h1 h3
h3 -> X h2
*** Results: 33% dropped (4/6 received)
```

Iperf

```
mininet> h3 iperf -s -u &
0.0.0.3 -u[1] 1451
mininet> h2 iperf -c 10.0.0.3 -u
-----
Client connecting to 10.0.0.3, UDP port 5001
Sending 1470 byte datagrams, IPG target: 11215.21 us (kalman adjust)
UDP buffer size: 208 KByte (default)
-----
[ 1] local 10.0.0.2 port 57472 connected with 10.0.0.3 port 5001
[ ID] Interval      Transfer      Bandwidth
[ 1] 0.0000-10.0155 sec 1.25 MBytes  1.05 Mbits/sec
[ 1] Sent 896 datagrams
[ 1] Server Report:
[ ID] Interval      Transfer      Bandwidth      Jitter    Lost/Total Datagrams
[ 1] 0.0000-10.0142 sec 1.25 MBytes  1.05 Mbits/sec  0.011 ms  0/895 (0%)
```

Flow_table

```
vishnu@LAPTOP-U8CDB0IM:~$ sudo ovs-ofctl dump-flows s1
[sudo] password for vishnu:
cookie=0x0, duration=179.609s, table=0, n_packets=2, n_bytes=196, priority=300,ip,nw_src=10.0.0.1,nw_dst=10.0.0.3 actions=drop
```

Controller_logs

```
vishnu_@LAPTOP-U0CDB0IM:~/pox$ cd ~/pox
./pox.py forwarding.l2_learning arp_controller
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
INFO:arp_controller:Blocking Controller Started
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.12.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
INFO:openflow.of_01:[00-00-00-00-00-01 1] connected
INFO:arp_controller:DROP rule installed for h1 -> h3
INFO:openflow.of_01:[00-00-00-00-00-01 1] closed
```

```
mininet> h1 ping -c 1 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
From 10.0.0.1 icmp_seq=1 Destination Host Unreachable

--- 10.0.0.2 ping statistics ---
1 packets transmitted, 0 received, +1 errors, 100% packet loss, time 0ms

mininet> h1 arp -n
Address                  HWtype  HWaddress          Flags Mask          Iface
10.0.0.2                  *        (incomplete)              h1-eth0
```